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Epidemiology of quadriceps muscle strain injuries in elite male Australian football players

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Abstract

Objective: To describe the epidemiology of quadriceps muscle strain injury (QMSI) in elite Australian Football League (AFL) players, explore recovery milestones and determine whether recovery is impacted by factors such as injury type (index vs. re-injury), the primary muscle injured and the mechanism of injury.

Measures: All QMSI data reported to the Soft Tissue Injury Registry of the AFL from the 2014 to 2020 seasons were evaluated. Player demographic data, circumstances of injury, MRI reports and recovery outcomes following injury were extracted. Descriptive statistics and frequency distributions are presented. Recovery outcomes for injury type, primary muscle injured and the mechanism of injury were compared using univariate analyses.

Results: There were 164 QMSIs from 122 players reported (134 index; 30 re-injuries). Almost all (91.3%) QMSIs involved the rectus femoris. Half (48.4%) of the QMSIs occurred during kicking and most commonly affected the dominant kicking leg (72%). The majority occurred at training (64.6%). All re-injuries involved the rectus femoris, most occurred from kicking (63.0%) and within 6 months of the preceding injury (70%). The mean return to play (RTP) time was 25.4 days (95%CI = 22.6–28.2) and rectus femoris injuries took around 14 days longer to RTP than vastii injuries ($p = 0.001$). QMSIs with a kicking mechanism took the longest to RTP of all injury mechanisms.

Conclusion: In AFL players, QMSIs occur mostly in the dominant leg from a kicking mechanism. Rectus femoris injuries are more prevalent and result in longer RTP time frames. Re-injuries exclusively involved the rectus femoris, primarily from kicking.

KEYWORDS

football, muscle injury, muscle strain, prognosis, quadriceps, sport

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1 | INTRODUCTION

Quadricep muscle strain injuries (QMSIs) are common in sports.[1] Athletes from sports involving kicking and sprinting are especially impacted by QMSI.[2–5] A decade of injury surveillance data from the Australian Football League (AFL) highlight incidence rates of 1–2 new QMSIs per team per season, resulting in an average time loss of 4.4 games.[2] The annual recurrence rate for QMSI averaged 8% and reached as high as 19% during this 10-year period,[2] which is comparable to re-injury rates observed in elite soccer.[4]

Australian Football is a fast-paced team sport that requires athletes to complete a combination of high-speed running, sprinting, acceleration, and cutting, as well as technical skills including handballs, kicking, tackling, and jumping[6,7] Over the course of a match, players typically run more than 11–13.5 km including ≈3 km of high-speed running (>4.17 m/s^2), peak sprint speeds of >8 m/s, and more than 80 explosive accelerations (>2.78 m/s^2).[8–11] Additionally, kicking is an essential skill[12,13] with the ability to kick with accuracy over long distances and successfully score goals vital to team success.[14] Elite AFL players may kick the ball up to ≈20 times per match, while kicking workloads are substantially greater during training activities.[15]

Given this repeated exposure to kicking and high speed running, the quadriceps muscles of elite AFL players, particularly the rectus femoris, are exposed to significant stress,[16] which might explain the consistent prevalence of QMSI in the AFL. Despite this, data about the epidemiology and recovery time frames for QMSI is historically scarce compared to other common muscle strain injuries such as the hamstrings and calf muscles.[17–20] Information about the circumstances and clinical factors that may impact the occurrence and prognosis of QMSI would be valuable for clinicians planning rehabilitation and return to play (RTP).

Evidence is limited regarding the potential impact that player characteristics, injury location or mechanism of injury may have on recovery time frames and re-injury risk after QMSI. The aims of this study were to: (1) describe the epidemiology of QMSI in elite AFL players, including characteristics of injured players, cumulative incidence of injury (per season), injury circumstances, and recovery time frames after injury, and (2) evaluate whether clinical factors such as injury type (index vs. re-injury), primary muscle injured (rectus femoris vs. vastii) and mechanism of injury (kicking vs. running related activity vs. other activity) are associated with recovery time frames after QMSI.

2 | MATERIALS AND METHODS

Reporting of results followed the football-specific extension of the International Olympic Committee (IOC) Consensus Statement: Methods for Recording and Reporting of Epidemiological Data on Injury and Illness in Sport 2020[21] where possible.

Data were obtained from the Soft Tissue injury Registry of the AFL (STRAFL) for QMSIs that occurred during seven consecutive seasons (2014–2020). Data were gathered from all 18 AFL teams (range from 13 to 18 teams participating per year) (Data S1). Injuries were reported to the STRAFL when a club doctor and/or physiotherapist clinically diagnosed a player with a QMSI that resulted in time loss; either modification of training time or time loss from competition. Injuries caused by blunt trauma (i.e., contusions) were not included. A standardized injury report was completed by club physiotherapists, with clinical data for QMSIs recorded in the STRAFL in the following categories:

a. Player demographics; chronological age, height, weight, body mass index (BMI), playing age (number of career games prior to injury), ethnicity (indigenous, non-indigenous), leg dominance (preferred kicking leg), lower limb injury history (prior QMSI/hamstring muscle strain injury, hip/knee joint injury), and primary position played (forward, midfield, back, ruck, and utility)
b. Training and match history: stage of the season and whether or not the player had completed the majority of preseason (completed >80% of the prescribed preseason training program).
c. Circumstances of injury: date of injury, activity when the injury occurred (training, match, and other), timing of the injury within the match and the mechanism of injury (acceleration, deceleration, high intensity running, steady state running, sudden change of direction, kicking (kicking leg), kicking (stance leg), landing, jumping, gradual onset, nil specific, other).
d. Recovery outcomes following injury: number of days to reach pain free walking, number of days to return to running at >90% of maximum speed, number of days to return to full training and number of days to RTP.

Recovery data were incomplete for some cases for reasons including: the end of the season occurred; the player sustained a re-injury or a different injury during rehabilitation; the player was concurrently rehabilitating a separate injury; no matches were scheduled at the time of RTP (e.g., pre-season injuries); or no data were provided. Where re-injury occurred within the rehabilitation period, data were only recorded for recovery milestones prior to the point of re-injury.

Ethics approval was obtained from the AFL Research Board and La Trobe University Human Ethics Committees (FHEC13/057). Written consent was received from all players included in the study.

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3 | DATA HANDLING AND ANALYSIS

3.1 | Classification of index and re-injuries

Index injuries were defined as the first recorded QMSI for the injured side. QMSIs where no previous QMSI had occurred in the preceding 2 years were also classified as index injuries.

Re-injury was defined as subsequent QMSI that occurred on the same side and involved the same primary muscle location as the immediately preceding QMSI up to 2 years following the date of the prior injury. To identify re-injuries, recorded data for date of injury, side of injury and past history were assessed, while MRI reports were reviewed to confirm date and injury location. In the first 2 years of data collection, two injuries were classified as re-injuries due to details of a prior QMSI (with date and location of injury) being provided in the past history section of the players record.

3.2 | MRI assessment

Where possible, electronic copies of MRI reports (completed by radiologists at the clinic where the MRI was performed) were sourced to determine whether the injury predominantly occurred in the rectus femoris or vastii muscles. Where MRI was performed and the radiological report was available, the location of primary muscle injury was extracted by SP and TP. No MRI images were accessed or evaluated in this study. Cases were excluded from analyses related to the primary muscle injured where MRI reports were unavailable. Cases without MRI reports were included in the analysis of injury type, mechanism of injury, leg dominance and injury activity, where the primary location of injury was not a dependent variable.

3.3 | Statistics

Descriptive frequency distributions were generated to describe characteristics of the injured players; age (years), playing age (games), seasons in the AFL, height, weight, body mass index (BMI), position played, and ethnicity were recorded. Descriptive statistics were used to determine the number of injuries per season, the muscle injured, the stage of the season that the injury occurred (month), training and match histories at time of injury, injury mechanisms and the involvement of the dominant versus non-dominant legs. Data were also explored to identify the cumulative incidence of injury (per season) and of re-injury. Continuous variables were assessed for normality (Shapiro Wilk test), and outliers were removed (data 3*IQR below Q1/above Q3, or there was clinical inconsistency with results [Data S2]).

Most data were non-normally distributed. Mann–Whitney *U*-tests were used to evaluate between group differences in the time to reach recovery milestones for: (i) injury type (index vs. re-injury); (ii) the primary muscle injured (rectus femoris vs. vastii); (iii) leg dominance (dominant vs. non-dominant kicking leg) and; (iv) activity (training vs. match based injury). Effect size (r) calculations used the formula [$r = z/\sqrt{n}$][22] and were categorized as small ($r \leq 0.3$), medium ($r = 0.3$–0.5), or large ($r \geq 0.50$).[23] Kruskal–Wallis tests analyzed whether recovery milestones differed for alternative injury mechanisms (kicking vs. running related vs. all other mechanisms). Effect sizes were calculated for these comparisons: epsilon-squared estimate of effect size [$E^2_R = H/(n^2\text{-}1)/(n+1)$] where the coefficient assumes the value from 0 (indicating no relationship) to 1 (indicating a perfect relationship).[22] Statistical software (IBM SPSS Statistics 28.0; significance level $p < 0.05$) was used to perform data analyses.

4 | RESULTS

4.1 | Player demographics

A total of 122 different players sustained 164 separate QMSIs, reported to the STRAFL between November 2013 and September 2020 (Table 1). The mean chronological age at the time of each injury was 24.2 years (SD = 3.9, range 18.1–32.8), with a playing age of 70.3 games (SD = 79.6, range 0–308). Injuries affected players from all playing positions: forwards (25%), midfielders (29%), backs (36%), utilities (5%), and rucks (5%). Compared to players who sustained vastii injuries, those who sustained rectus femoris injuries were significantly shorter ($U = 370$, $r = 0.22$, $p = 0.01$) and lighter ($U = 356$, $r = 0.23$, $p = 0.01$).

4.2 | Incidence and type of quadriceps muscle strain injury

There were on average 23.4 QMSIs (index and re-injuries) per calendar year (range 16–31) across the competition (1.5 QMSIs per participating club per year). Of the 164 QMSIs, 134 were index injuries (81.7% of all QMSIs; 19.1 per calendar year), and 30 were re-injuries (18.3% of all QMSIs; 4.3 per calendar year) (Table 2).

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TABLE 1 Demographics of injured players.

	Age (years) (95% CI)	Height (cm) (95% CI)	Weight (kg) (95% CI)	Playing age (games played) (95% CI)	Years on AFL list (years) (95% CI)	Side injured (Dom./Non-Dom.)	Previous lower limb injury (QMSI/HS/Hip/Knee)	Playing position (Mid/Back/Fwd/Ut/Ruck)
All QMSI	24.2 (23.6–24.8)	187.2 (186.1–188.2)	85.6 (84.3–86.9)	70.3 (58.0–82.6)	4.8 (4.2–5.5)	72%/28%	50%/47%/25%/42.7%	29.3%/36%/25%/4.9%/4.9%
Injury type								
Index	23.9 (23.3–24.6)	187.3 (186.1–188.5)	85.6 (84.1–87.1)	63.7 (51.0–76.4)	4.6 (3.9–5.3)	70.1%/29.9%	38.8%/44.3%/25.4%/48%	28.4%/35.8%/24.6%/6%/5.2%
Re-injury	25.4 (23.9–27)	186.6 (184.2–189.1)	85.2 (82.8–87.7)	99.6 (64.0–135.2)	6.1 (4.5–7.7)	80%/20%	100%/63.3%/34.6%/37%	33.3%/36.7%/26.7%/0%/3.3%
Primary muscle injured								
Rectus femoris	24.5 (23.8–25.2)	186.7 (185.5–187.8)	86.5 (82.0–91.0)	76.1 (61.2–90.9)	5 (4.3–5.8)	73.8%/26.2%	52.4%/48%/21.7%/47.8%	28.6%/38.9%/24.6%/4.8%/3.2%
Vastii	24.6 (22.2–27.0)	191.8 (188.2–195.5)*	92 (87.0–97.0)*	67.8 (21.2–114.4)	5.5 (2.7–8.2)	63.6%/36.4%	54.5%/45.5%/63.6%/45.5%	9.1%/54.5%/9.1%/ 9.1%/18.2%

Abbreviations: Back, backline; Dom, dominant; Fwd, forward line; HS, hamstring; Mid, midfield; QMSI, quadricep muscle strain injury; Ut, utility.
*Statistically significant difference compared with rectus femoris injuries ($p < 0.05$).

4.3 | Primary muscle injured

MRI reports were available for 138 (84.1%) QMSIs. One of these injuries had no evidence of QMSI on MRI and was excluded, resulting in 137 QMSIs with a record of primary muscle injured. MRI reports were not available for 26 (15.9%) of cases because: (1) MRI was not performed, or, (2) MRI was performed but the report was not available. The majority of QMSIs occurred in the rectus femoris ($n = 126$, 92%). 11 (8%) occurred in the vastii. All re-injuries ($n = 30$) had MRI reports and occurred exclusively in the rectus femoris.

4.4 | Re-injury

Re-injuries occurred at a mean of 127.1 days (SD: 146.6, median: 53, range: 8–566) following the date of the preceding QMSI. Approximately one quarter (26.7%) of re-injuries occurred prior to the mean RTP time for index QMSIs (24.2 days). Re-injury occurred before RTP in six cases (exacerbations[21]) and on the day of RTP in two cases. Approximately three quarters (77%) of re-injuries occurred within the same season as the preceding QMSI (Figure 1).

4.5 | Timing of injury

Most QMSIs occurred during the competitive season (March to September: 73%, $n = 120$). The frequency of QMSIs was highest in the middle months of the season (May: 15.2%, June: 12.2%, July: 16.5%). The frequency of re-injury was highest late in the competitive season (July: 23.3%, August: 20%) (Data S3). QMSIs occurred during official matches (30.5%), training (64.6%), and other activities (e.g., gym and warm up) (4.9%). The proportion of match vs. training injuries was similar for index injuries (matches: 29.1%, training: 64.9%), and re-injuries (matches: 36.7%, training: 63.3%) (Table 3). Match related QMSIs (i.e., cases that presented during match day activities or post-match) occurred in the warm up (2.4%), the first half (47.6%), the second half (40.5%), and post-match (9.5%) (Data S4).

4.6 | Pre-season completeness

A similar proportion of players sustaining QMSI had completed ≥80% of preseason compared those who had not (44.5% vs. 55.5%, $n = 146$), while proportions were also similar between players sustaining an index injury (≥80% preseason: 45.5%) and those with a re-injury (≥80% preseason: 40%).

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TABLE 2 Injuries per season by type and primary muscle injured.

	2014	2015	2016	2017	2018	2019	2020	Total
TOTAL QMSI	20	18	16	24	26	29	31	164
Index	16	17	14	23	20	22	22	134 (81.7%)
Re-injury	4	1	2	1	6	7	9	30 (18.3%)
TOTAL QMSI WITH MRI	17	14	15	17	22	24	28	137 (83.5%)
Rectus femoris	16	11	14	16	19	22	28	126 (92%)
Vastii	1	3	1	1	3	2	0	11 (8%)

Abbreviations: MRI, magnetic resonance imaging; QMSI, quadricep muscle strain injury.

4.7 | Mechanism of injury

Almost half (48.4%) of QMSIs occurred when kicking (kicking leg). Running-related mechanisms accounted for 31% of QMSIs, with acceleration (14.8%) and high intensity running (9%) being the most common running related injury mechanisms. A small proportion (12.9%) of QMSIs lacked a specific inciting mechanism (Figure 2). Index QMSIs commonly occurred when kicking (45%). Kicking was the most common mechanism of re-injury (63%). Similar percentages of kicking and running related injuries were seen in matches (kicking: 45%, running: 33%) and training (kicking; 51%, running: 32%). When considering only rectus femoris injuries ($N=120$), the distribution of injury mechanisms was similar (Data S5).

4.8 | Leg dominance

The majority of QMSIs affected the dominant leg (72%). Nearly three quarters (74%) of dominant leg QMSIs involved the rectus femoris. The dominant leg was injured in 70% of index QMSIs and 80% of re-injuries. Most dominant leg QMSIs occurred when kicking (61%), whereas kicking only accounted for 16% of non-dominant leg QMSIs. Injuries due to running-related activities were more evenly distributed between the dominant (54%) and non-dominant (46%) legs.

4.9 | Recovery post injury

There was large variation in the number of days to reach recovery milestones after QMSI and the number of completed data points for each recovery milestone varied (Table 4). For all QMSIs, players took on average 2.5 ± 2.3 days to return to pain free walking (median: 2, range 0–13), 19 ± 13.4 days to return to sprinting at >90% of maximum speed (median: 16, range: 3–70), 23.5 ± 15.3 days to return to full training (median: 21, range: 1–83) and 25.4 ± 14.8 days to RTP (median: 23, range: 5–82).

Rectus femoris injuries took longer to return to sprinting at >90% speed ($U=233.5$, $r=0.27$, $p=0.002$), return to full training ($U=271.5$, $r=0.3$, $p=0.005$) and RTP ($U=152$, $r=0.34$, $p=0.001$) compared to vastii injuries. QMSIs occurring from kicking took longer to return to sprinting at >90% speed ($H=6.849$ (2), $E^2_R=0.05$, $p=0.03$) and longer to RTP ($H=8.185$ (2), $E^2_R=0.08$, $p=0.02$) compared to QMSIs with non running-related mechanisms (e.g., landing, jumping, gradual onset, nil specific, other). No other comparisons of interest reached significance despite variation in the time taken to RTP for: (i) re-injuries (+5.7 days) versus index injuries, (ii) training (+2.3 days) versus match injuries, (iii) kicking (+2.7 days) versus running-related mechanisms and (iv) QMSIs affecting the dominant (−0.8 days) versus non-dominant leg ($p>0.05$ for all).

5 | DISCUSSION

This study identified four key findings from the analysis of 164 unique QMSIs reported to the STRAFL across seven consecutive seasons. First, the majority of MRI-confirmed QMSIs occurred in the rectus femoris, and re-injuries exclusively involved rectus femoris. Second, kicking was the most common mechanism of injury for QMSI, with injuries most frequently occurring in the dominant kicking leg. Third, contrary to evidence in other common muscle strain injuries, there were no differences in the player demographics or recovery time frames between index QMSIs and re-injuries. Fourth, recovery after QMSI differed depending on the primary muscle injured (rectus femoris vs. vastii) and the mechanism of injury (kicking vs. running-related mechanism vs. other mechanism).

Recovery time frames for QMSIs reported in this study are comparable to other sports, with similar RTP outcomes described in elite soccer (19.5 days (95% CI: 18.1–20.9).[4] Return to full training time frames for QMSIs involving

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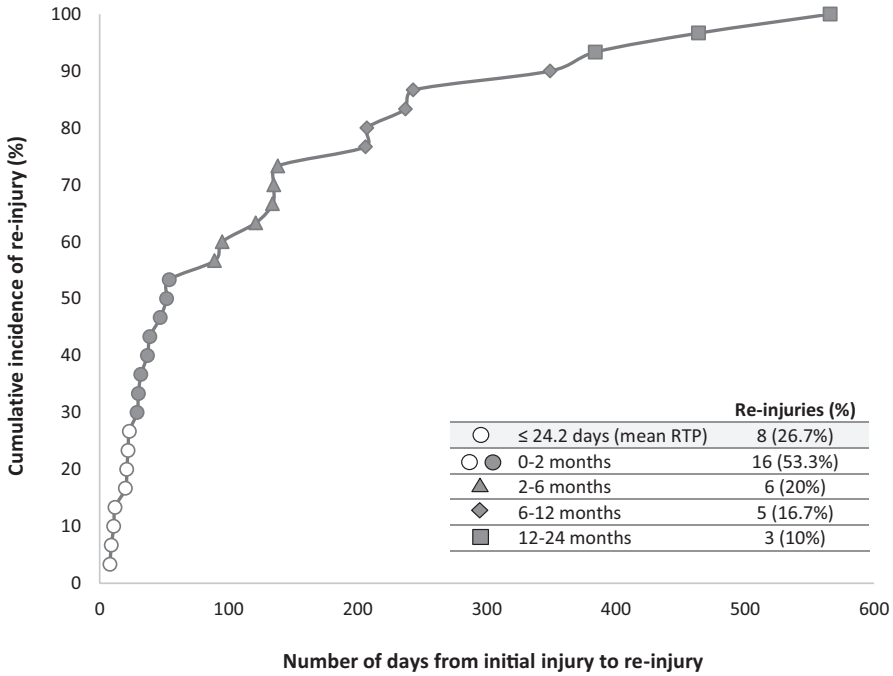

FIGURE 1 Cumulative incidence of re-injury.

TABLE 3 Match and training related quadriceps muscle strain injuries.

	Match (%)	Training (%)	Other (%)
All QMSI ($N=164$)	50 (30.5)	106 (64.6)	8 (4.9)
Injury type			
Index ($N=134$)	39 (29.1)	87 (64.9)	8 (6)
Re-injury ($N=30$)	11 (36.7)	19 (63.3)	0 (0)
Injury location			
Rectus femoris ($N=126$)	39 (40)	80 (63.5)	7 (5.5)
Vastii ($N=11$)	4 (36.4)	6 (54.5)	1 (9.1)

Abbreviation: QMSI, quadricep muscle strain injury.

rectus femoris are also consistent with data from elite level sprinters (20.4 ± 14.8 days),[24] and an earlier study of a small sample of 25 AFL players (18.6 ± 12.3 days).[25]

In our analysis, rectus femoris injuries were more prevalent than vastii injuries. Although there were a limited number of vastii injuries ($n=11$) from which to make comparisons, the time frames to return to sprinting, full training and RTP were longer for rectus femoris injuries, which is consistent with previous research in a similar population.[25] These findings may be unsurprising given the differences in the anatomical structure, mechanical function and load requirements of these muscles.[25,26] During athletic activities such as sprinting and kicking, the rectus femoris undergoes large and rapid length changes during its active stretch-shortening cycles due to its biarticular structure,[1,26] which may explain its propensity for injury.

Kicking (kicking leg) was the most common mechanism for QMSIs in this study. Our results are consistent with findings in other sports, with 53% of kicking injuries in rugby union players occurring in the quadriceps[27] and 54% of rectus femoris injuries in soccer players occurring from a kicking mechanism.[28] During the most common kick used in Australian football, the drop punt, the rectus femoris of the kicking leg must moderate high eccentric forces through the wind-up phase (hip extension and knee flexion), before contracting concentrically during forward swing to rapidly accelerate the leg towards the ball (hip flexion and knee extension).[13,29] This action is similar to the instep soccer kick where peak rectus femoris activation occurs in the wind-up phase prior to ball contact.[30] This represents a potential time when the muscle is at risk, although injury occurring at ball strike when the ball creates a retarding torque to concentric knee extension, is typically reported.[31] The higher frequency of QMSI occurring in training in our cohort (opposite to calf[17] and hamstring injuries[32,33] that occur more frequently in matches) may be a consequence of the higher kicking loads encountered at training compared to matches in Australian football.[15]

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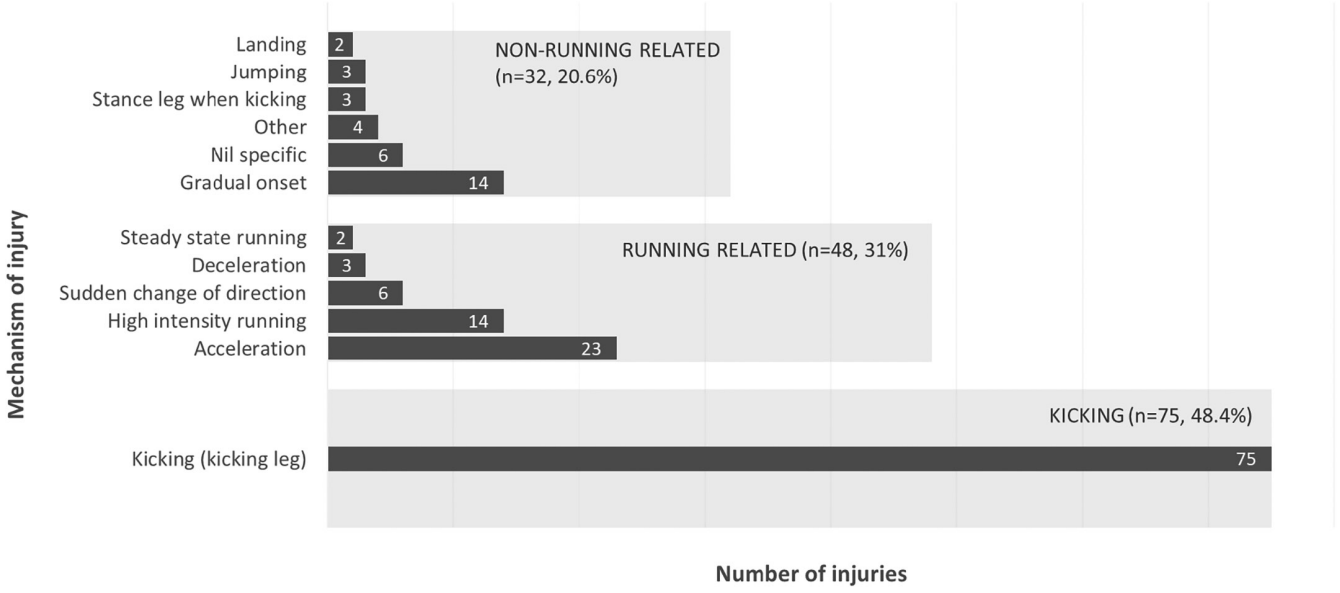

FIGURE 2 Mechanism of injury grouped by kicking, running and other non-running related mechanisms. Data is presented for number of injuries per mechanism and proportion of total injuries.

Sprinting or running-related injury mechanisms accounted for 31% of all QMSIs and were more frequently associated with acceleration and high intensity mechanisms. Similar findings have been reported in soccer, with 30.4% of rectus femoris injuries occurring from a sprinting mechanism,[28] and in elite track and field athletes where rectus femoris injuries occur when sprinting.[24] Electromyographic studies of sprinting have found the rectus femoris to be highly activated during the late stance and early swing period of the stride cycle.[34,35] The loads experienced by the rectus femoris appear to increase dramatically with faster running, with predicted peak forces during early swing to increase 2.4-fold when running speed progressed from ≈5 m/s to maximal sprinting.[36]

The mechanism of injury can influence the length of recovery following muscle strain injuries.[17,37–39] In our study, RTP time frames were longer for QMSIs with a kicking (kicking leg) mechanism when compared to gradual onset/other mechanisms (e.g., landing, jumping), and slightly longer than QMSIs due to running related mechanisms (although not statistically significant). Kicking-related QMSIs may involve different anatomical locations and patterns of tissue damage within the rectus femoris compared to running related injuries. In soccer players there is an association between proximal tendon injury location and a kicking mechanism.[28] It is possible that kicking related injuries more frequently involve the connective tissue elements within the rectus femoris, the integrity of which are critical to allow the muscle-tendon unit to develop high forces during ballistic activities such as kicking.[12,28] The location of injury in the rectus femoris, especially cases involving proximal connective tissue elements, may require longer rehabilitation periods to allow adequate tissue healing[40] and mitigate the risk of re-injury,[25,41,42] but further research is warranted.

Re-injuries occurred exclusively in the rectus femoris, with a re-injury rate (18%) comparable to elite soccer (15.6%).[4] A high early cumulative incidence of re-injury (53% within 2 months of preceding injury) is similar to data for calf (54% under 2 months from index injury)[17] and hamstring (50% within 50 days of index injury)[43] muscle strain injuries. The etiology of early muscle strain re-injuries is multifactorial.[44,45] Exposure of healing tissue to loads that surpass capacity, at a time when it is less capable of adaptation, is likely to be a factor.[46] This may be especially relevant in elite sport where contextual pressures exist to RTP and return to performance[47] as soon as possible. Re-injuries may occur due to muscle inhibition,[48] scarring[49] and architectural changes[50] that may be present in previously injured muscle, even when clinical signs may have resolved. Longer average RTP times for re-injuries of QMSI have been shown in soccer (+4.3 days),[4] and while our data are comparable and appear to show a clinically relevant difference (+5.7 days), this comparison was not statistically significant. Incomplete outcome data for re-injury cases may have impacted these results, and further analysis may provide more insight.

6 | LIMITATIONS

The STRAFL is designed to collect data on injured players. As a result, comparisons of characteristics between players with and without QMSI is not possible. Exposure data

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TABLE 4 Time to reach recovery milestones following quadriceps muscle strain injury (days).

	Pain free walking				Return to running >90% max speed				Return to full training				Return to play			
	N	Mean (SD)	Median (IQR)	95% CI	*N*	Mean (SD)	Median (IQR)	95% CI	*N*	Mean (SD)	Median (IQR)	95% CI	*N*	Mean (SD)	Median (IQR)	95% CI
All QMSI	152	2.5 (2.3)	2 (2)	2.1–2.8	137	19 (13.4)	16 (16)	16.7–21.3	137	23.5 (15.3)	21 (19)	20.9–26.1	110	25.4 (14.8)	23 (18)	22.6–28.2
Injury type																
Index	127	2.5 (2.4)	2 (2)	2.1–2.9	115	18.6 (13.6)	15 (17)	16–21.1	116	22.7 (15)	21 (17)	20–25.5	93	24.5 (14.9)	22 (17)	21.5–27.6
Re-injury	25	2.4 (1.9)	2 (3)	1.6–3.2	22	21.4 (11.9)	20.5 (11)	16.1–26.7	21	27.6 (16.6)	25 (14)	20.1–35.2	17	30.2 (13.9)	28 (17)	23–37.3
Primary muscle injured																
Rectus femoris	115	2.6 (2.5)	2 (3)	2.1–3.1	101	21.4 (13.9)	19 (14)	18.7–24.2	103	25.6 (15.7)	24 (18)	22.5–28.7	80	28.8 (15.1)	28 (15)	25.4–32.1
Vastii	11	2.2 (1.9)	2 (4)	0.9–3.5	11	11.4 (12.3)*	6 (5)	3.1–19.7	11	13.8 (12.6)*	7 (15)	5.3–22.3	10	14.9 (11.1)*	11.5 (11)	7–22.8
Mechanism of injury																
Kicking	70	2.5 (2.5)	2 (2)	1.9–3	63	19.9 (12.9)	19 (17)	16.7–23.1	64	23.5 (15.3)	22 (17)	19.7–27.3	50	28 (15.7)	28 (20)	23.5–32.4
Running	46	2.8 (2.2)	2 (2)	2.1–3.4	40	20.7 (14.3)	16 (16)	16.1–25.2	40	24.7 (14.6)	21 (15)	20–29.4	33	25.3 (14.1)	22 (13)	20.35–30.32
Non-running	31	2.1 (2.3)	2 (2)	1.3–3	28	14.9 (13.7)**	11.5 (11)	9.6–20.2	28	20.8 (17.1)	16 (22)	14.1–27.4	22	17.8 (10.2)**	14 (17)	13.3–22.3
Leg dominance																
Dominant	110	2.4 (2.2)	2 (2)	2–2.9	100	18.7 (12.2)	17 (16)	16.3–21.1	98	22.5 (14)	21 (17)	19.7–25.3	77	25.2 (14.3)	22 (20)	21.9–28.4
Non-dominant	42	2.6 (2.5)	2 (2)	1.8–3.4	37	19.8 (16.4)	15 (18)	14.4–25.3	39	25.8 (18.3)	21 (21)	19.9–31.7	33	26 (16)	25 (17)	20.3–31.7
Injury activity																
Match	48	2.9 (2.4)	2 (3)	2.2–3.6	44	15.9 (11.3)	13.5 (14)	12.5–19.4	39	21.4 (17.1)	18 (16)	15.9–27	38	23.6 (14.7)	20.5 (20)	18.8–28.5
Training	97	2.2 (2.1)	2 (2)	1.8–2.6	85	19.7 (13.1)	18 (17)	16.9–22.6	90	23.5 (13.4)	21.5 (18)	20.7–26.3	66	25.9 (14.4)	25 (18)	22.3–29.4

Abbreviations: CI, confidence interval; IQR, interquartile range; Max, maximum; *N*=, number of complete data points for each recovery milestone per variable; SD, standard deviation.

*Statistically significant difference compared with rectus femoris injuries ($p < 0.05$). **Statistically significant difference compared with injuries with a kicking mechanism ($p < 0.05$).

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are not routinely collected, which prevents rates for training and match exposure to be reported. As the reporting of QMSI to the STRAFL was completed voluntarily, it is possible that not all injuries that occurred during the surveillance period were captured, especially for pre-season injuries where data reporting was variable between clubs. Data were not received from all clubs for every calendar year, and therefore the data presented in this study may underrepresent the true incidence of QMSI in the AFL. MRI findings relied on the accuracy of the reporting radiologists. MRI reports were unavailable for a small proportion of QMSIs, which excluded some cases from analyses related to the primary muscle injured. Variation in recovery milestone data reporting affected data availability for some analyses (e.g., missing RTP data for QMSIs that occurred late in the season) and thus our results should be interpreted accordingly. Data were only recorded for elite male AFL players, and therefore our findings may lack generalizability to female athletes, sub-elite Australian football, or to athletes from other sports.

7 | PERSPECTIVES

QMSIs are prevalent in the AFL. Despite this, little is known about the epidemiology and recovery time frames for QMSIs when compared to other common muscle injuries in football. In our study, rectus femoris injuries represent the most significant injury burden for QMSI in professional Australian Football, with all re-injuries isolated to the rectus femoris. The highest proportion of QMSIs occurred from a kicking mechanism and affected the dominant kicking leg. Longer RTP timeframes may be expected for rectus femoris injuries compared to vastii injuries, and for QMSI with a kicking mechanism compared to injuries from non-running related mechanisms such as landing and jumping or a gradual onset. Improved knowledge of the clinical factors that impact outcomes for recovery and re-injury after QMSI, can guide injury management and prevention.

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The authors have no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

No data are available. Data are confidential and not available upon request.

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